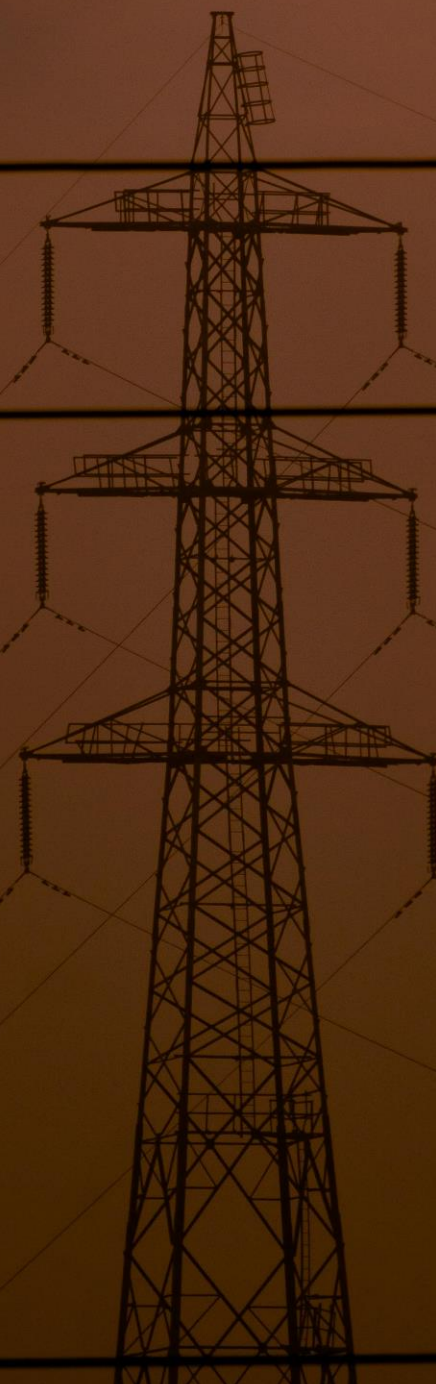




NITI Aayog



SECTOR AT A GLANCE 01

Energy Sector in India

An Overview

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SECTOR AT A GLANCE 01

ENERGY SECTOR IN INDIA

AN OVERVIEW

The **'Sector-at-a-Glance'** provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

India has witnessed a substantial surge in energy demand due to population growth, urbanization, industrial expansion, and economic development. Despite the increase in energy consumption from 700 Mtoe in 2010 to 880 Mtoe in 2020, per capita energy consumption remains low, constituting one-third of the global average.¹ Electricity access has notably improved, from 76% in 2010 to 99% in 2020.² Over 80% of India's energy needs are met by coal, oil, and traditional biomass, with efforts underway to diversify and increase the share of renewables.³ In the last decade, modern renewables saw the most significant increase in its share, followed by coal and oil, while the shares of traditional biomass and natural gas decreased.⁴ With a commitment to achieving net-zero targets by 2070, India has set ambitious goals for 2030, including installing 500 GW of renewable energy capacity and reducing emissions intensity.⁵

With the right support for India's energy ecosystem, it has the potential to ensure reliable access to affordable energy for all its citizens, improved energy security and independence, greater sustainability, and higher economic growth.

However, India's energy sector faces a range of challenges, including an over-reliance on fossil-based energy sources, high energy prices, underutilization of existing installed capacity, poor electricity reliability, dependency on imports for oil and natural gas, and the challenge of balancing an increase in energy demand while meeting decarbonization targets.

Given these opportunities and challenges, increasing demand for electrification, diversifying energy supplies, expanding the country's exploration and production activities, developing and promoting clean coal technologies, adopting alternative energy sources, transitioning towards a gas-based economy, and promoting the use of green hydrogen and electric vehicles are the major policy focus areas. Several ongoing and recent central government initiatives, including but not limited to the National Solar Mission, the Energy Conservation (Amendment) Act, the National Policy on Bio-fuels, and the Green Energy Corridor focus on sustainably meeting India's increasing energy needs.

Promoting renewable deployment and infrastructure, advancing energy security through greater and cleaner domestic production of fossil fuels, reforming India's power and electricity markets, accelerating energy storage and efficiency solutions, ensuring energy access for all, and promoting integrated energy policymaking are critical efforts to continue advancing India's energy sector. This note presents key trends related to different components of energy in India and actions for the way forward.

Figure 1: India's energy demand and energy supply, 2022

Energy demand in India



Source: NITI Aayog, India Energy Security Scenario (as under Least Effort Scenario – All Level 1), accessed December 2023

1 IEA, Total primary energy demand in India, 2000-2020, 2021; Ministry of Finance, India need to increase its per capita energy consumption at least 2.5 times to increase its Real Per Capita GDP by \$5000 per capita to enter the Upper-middle Income Group, 2019

2 World Bank, Access to electricity (% of population) – India, accessed June 2023; IEA, India's clean energy transition is rapidly underway, benefiting the entire world, 2022

3 Modern renewable sources refers to all renewable sources except for traditional biomass, this will include solar, wind, hydropower, bioenergy ; IEA, Total primary energy demand in India, 2000-2020, 2021

4 Modern renewable sources refers to all renewable sources except for traditional biomass, this will include solar, wind, hydropower, bioenergy ; IEA, Total primary energy demand in India, 2000-2020, 2021

5 Ministry of Finance, Summary of The Union Budget 2023-24; Ministry of Finance, Ministry of Petroleum and Natural Gas- Demand No. 76, 2023-2024

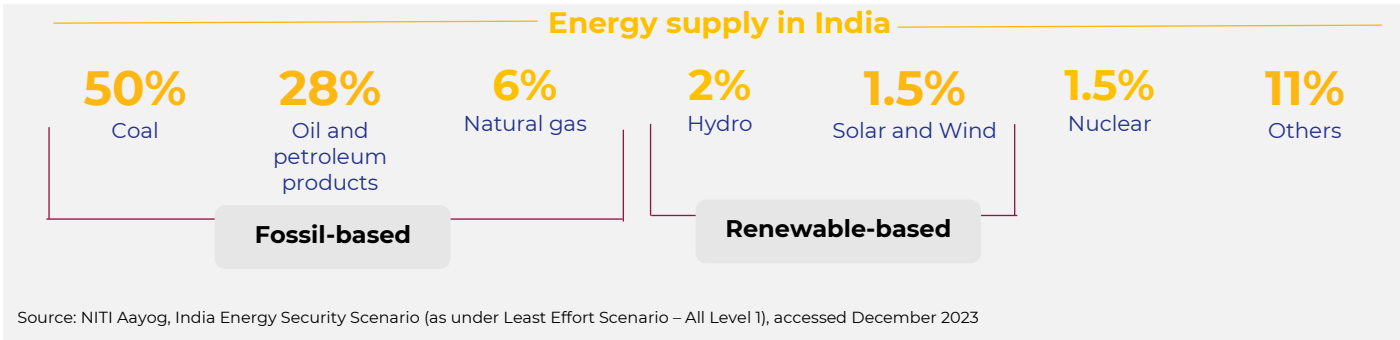
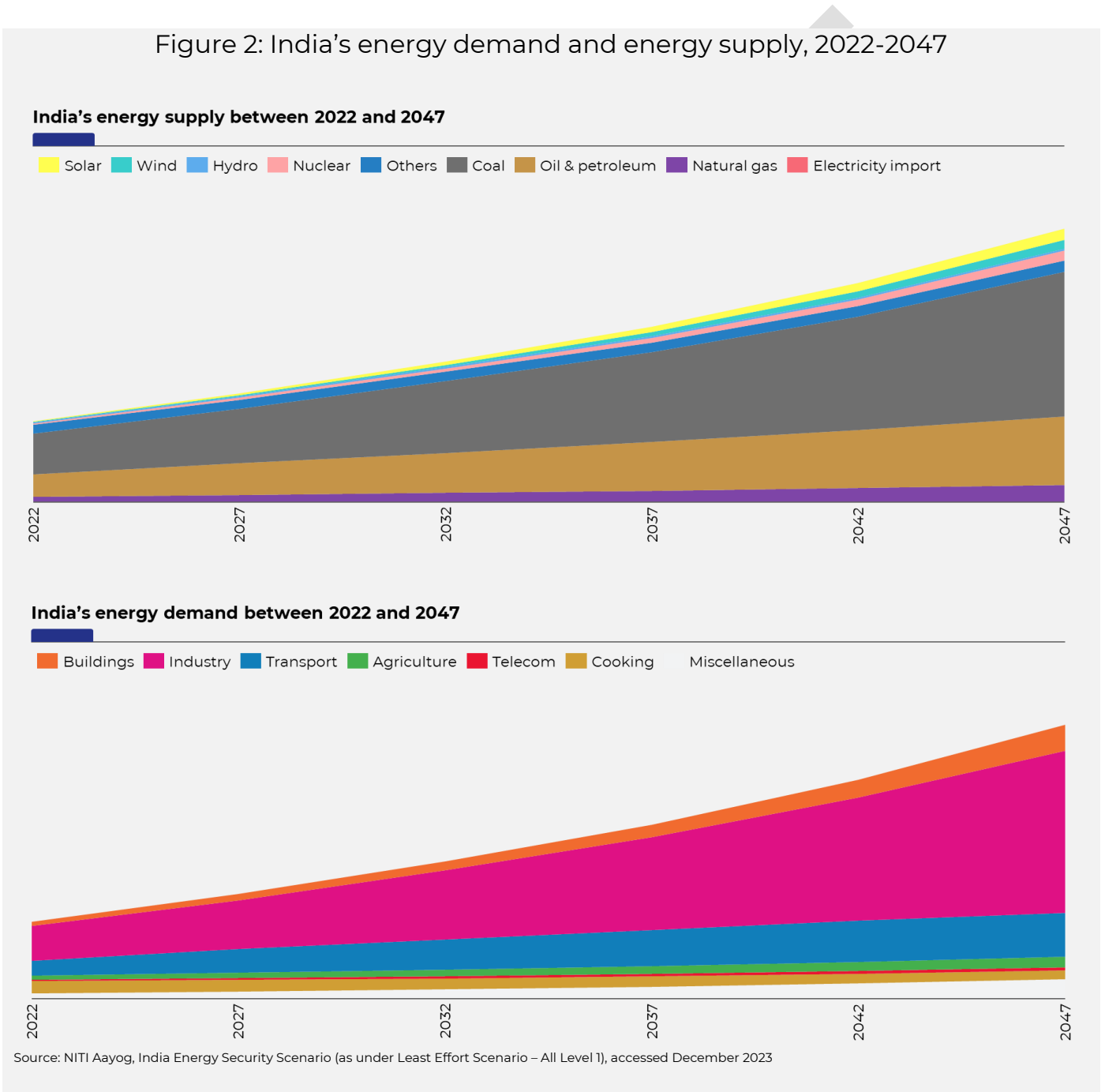


Figure 2: India’s energy demand and energy supply, 2022-2047



Sub-sectoral trends

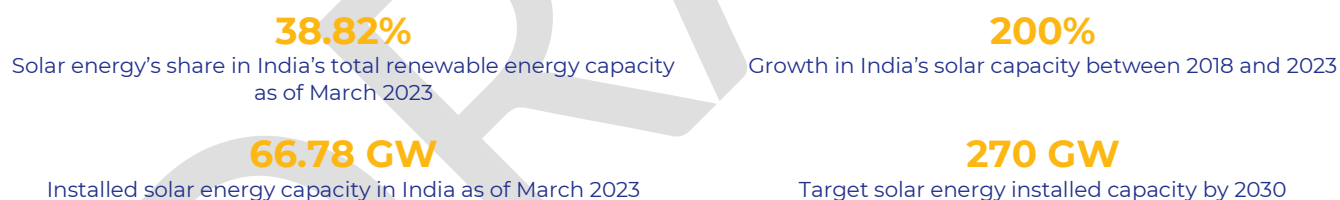
This section presents a summary view of all the relevant statistics and trends related to the following sub-sectors within the energy sector:

- Solar Energy
- Wind Energy
- Bio Energy
- Hydrogen Energy
- Atomic Energy
- Hydropower
- Emerging Technologies
- Oil and Natural Gas
- Coal
- Electricity
- Integrated Energy Assessment
- Energy Storage
- Energy Efficiency
- Negative Emission Technologies

Solar Energy

Solar energy involves harnessing the power of sunlight and converting it into usable electricity or heat. The solar sector in India encompasses both utility-scale solar power plants and distributed solar systems, such as rooftop solar panels. The country has made remarkable strides in solar energy deployment, with ambitious targets and large-scale projects.

Figure: Key solar energy statistics



Source: Central Electricity Authority, Installed Capacity Report, Mar 2023, 2023; Ministry of New and Renewable Energy, Solar Energy capacity has nearly tripled in last 5 years from 21651 MW to 64380 MW, 2023

India's domestic PV manufacturing potential has more than doubled between 2020 and 2023:

India's solar module manufacturing capacity is expected to grow fourfold by 2025 compared to 2021.⁶ By 2026, India is projected to reach a manufacturing capacity of 110 GW, achieving self-sufficiency in meeting the demand for PV modules.⁷ Leading players in the PV manufacturing industry include Waaree, Adani Solar, and Vikram Solar.⁸

Commercial and industrial (C&I) consumers comprise a growing share of solar demand:

In the rooftop solar segment, 78% of all installations by C&I consumers in 2024.⁹ The top five states comprising 50% of total C&I rooftop installations in solar energy are Rajasthan, Gujarat, Maharashtra, Tamil Nadu, and Karnataka in 2019.¹⁰ Residential demand has not grown significantly, with less than 1% of

⁶ Ministry of Information and Broadcasting, Union Budget 2022-2023: India Embarks on a Solar Journey, 2022

⁷ IEEFA, India's photovoltaic manufacturing capacity set to surge, 2023

⁸ JMK and IEEFA, India's Photovoltaic Manufacturing Capacity Set to Surge, 2023

⁹ Bridge to India, India RE Navigator- Rooftop solar, accessed June 2023

¹⁰ JMK Research and IEEFA, Powering Up Sunshine – Untapped Opportunities in India's Rooftop Solar Market, 2020

households in urban India installing rooftop solar systems.¹¹ To raise this demand, the government launched the Pradhan Mantri Suryodaya Yojana in 2024 to install rooftop solar in 10 million households.¹²

In 2022-23, solar power projects generated approximately 82 billion kWh of electricity:

In India, power generation from solar energy increased by 64% from 2019-20 to 2022-23 (till January).¹³ By 2027, solar energy is projected to contribute to approximately 9% of the overall electricity generation (utility) and is expected to increase to around 19% by 2047.¹⁴

Wind Energy

India has made significant progress in enhancing its wind capacity. Wind energy contributed to 24.78% of the cumulative installed power generation capacity of Renewable Energy Sources (RES) in March 2023.¹⁵ India ranks as the fourth-largest market for onshore wind installations worldwide, contributing 5% to the total global onshore wind installations, and possesses significant potential for both onshore and offshore wind energy.¹⁶ To harness India's offshore wind potential, the government launched the National Offshore Wind Energy Policy in 2015 to establish offshore wind power projects by promoting the deployment of offshore wind farms in Exclusive Economic Zones.¹⁷

Figure: Key wind energy statistics



Source: Central Electricity Authority, Installed Capacity Report, Mar 2023, 2023; NITI Aayog, India Energy Security Scenarios (IESS) 2047, 2023; Ministry of New and Renewable Energy, Installed Wind Energy Capacity, 2023

India is the second-largest hub in the Asia-Pacific region for wind turbine manufacturing, contributing to 7% of the global wind turbine manufacturing capacity as of 2023¹⁸:

India primarily leads in the assembly of wind turbines and the production of essential components such as gearboxes, blades, and generators in the APAC region.¹⁹ The country's current annual capacity for manufacturing domestic wind turbines is approximately 15,000 MW.²⁰ Notably, the wind sector has achieved significant indigenisation by increasing domestic manufacturing of components which were earlier being imported, with around 70-80% reliance on domestic manufacturing.²¹

Growth rate for wind energy has declined in the past five years:

In the past five years, the rate of wind energy installations has decreased, with an average growth rate of about 5% between 2019-2023, compared to around 12% between 2012-2016.²² A slowdown was observed during 2017-2019, which can be attributed to multiple changes in wind policies, such as the introduction

¹¹ Bridge to India, India Residential Rooftop Solar Market 2023, 2023

¹² Prime Minister's Office, 1 crore households to get rooftop solar under Pradhan Mantri Suryodaya Yojana, 2024

¹³ Ministry of New and Renewable Energy, BU Solar Energy produced this financial year till January, 2023; SUPRABHA and NISE, National Online Rooftop Solar Data Monitoring Centre (NORS-DMC), 2019

¹⁴ NITI Aayog, India Energy Security Scenarios 2047, Accessed August 2023

¹⁵ Central Electricity Authority, Executive Summary on Power Sector, June 2023; Note: RES include Small Hydro Project, Biomass Power, Urban and Industrial Waste Power, Solar and Wind Energy

¹⁶ Ministry of New and Renewable Energy, India attains global 4th and 5th positions in wind and solar power installed capacities; India now at 5th global position for overall installed renewable energy capacity, 2018

¹⁷ Ministry of New and Renewable Energy, Wind Policies, 2015

¹⁸ Global Wind Energy Council, Global Wind Report, 2023

¹⁹ Global Wind Energy Council, Global Wind Report, 2023

²⁰ Ministry of New and Renewable Energy, Wind Overview, Accessed June 2023

²¹ Ministry of New and Renewable Energy, Wind Energy – Manufacturing, Accessed June 2023

²² Central Electricity Authority, All India Electricity Statistics, 2023; Ministry of Power, Power Sector at a Glance, 2023

of reverse bidding.²³ The slowdown in 2020-2021 can be attributed to the COVID-19 pandemic's first and second waves and disruptions in the wind supply chain during the monsoon season.²⁴

Wind sector to create 1 million jobs in the next three decades:

Over 25 years, the projected installation of 19.4 GW of onshore wind energy in India from 2022 to 2026 is anticipated to generate 1 million direct and indirect jobs. These jobs will span various sectors, including manufacturing, project development, construction, operations, and maintenance (O&M), and installation.²⁵

Bio Energy

Bioenergy, a renewable energy source, is derived from biomass and feedstock including agricultural residues, forestry residues, animal waste, and organic municipal solid waste. Through processes like combustion, anaerobic digestion and biochemical conversion, bioenergy is produced in the forms of Biopower (electricity), Biofuels (ethanol, sustainable aviation fuel, and biodiesel for transportation), Biogas and briquettes (cooking), and waste to energy.

Figure: Key bio energy statistics



Source: Ministry of Power, Power sector at a glance ALL India, accessed on December 2023; Ministry of New and Renewable Energy, Bio Energy, 2022

India's compressed biogas (CBG) potential is 62 million metric tonnes (MMT), and targets have been set to achieve 15 MMT of CBG by 2024-25:

As per the MNRE, India can produce 62 MMT of CBG if the full potential of compressed CBG is exploited which is sufficient to fulfil the whole natural gas demand of the country. India has set the target to set up 5,000 commercial plants to produce 15 MMT of CBG by 2024-25 under the Sustainable Alternative Towards Affordable Transportation (SATAT) scheme.²⁶

India's biopower generation capacity and biofuel production have experienced substantial growth:

Biopower generation capacity as of May 2023 stands at 10.2 GW, marking a ~2X increase from 4.83 GW in 2015-16.²⁷ Bioethanol production has increased ~20X from 15.4 crore litres in 2012-13 to 304 crore litres in 2020-21.²⁸ Additionally, the production of biodiesels increased ~2X from 10 crore litres in 2010-11 to 18.5 crore litres in 2022-23.²⁹

Government initiatives are instrumental in promoting bioenergy utilization for electricity:

Bio-electricity's contribution to the overall electricity mix has grown over the past decade, contributing to 3% of total electricity production as of 2019. The Amended National Tariff Policy-2016 mandates Distribution Licensee(s) to purchase 100% of the power generated by Waste-to-Energy plants in the state. This can benefit biogas plants as well.³⁰ Additionally, the (Amended) National Policy on Bio-Fuels-2022 has set a target of 20% blending of ethanol in petrol by Ethanol Supply Year (ESY) 2025-26.³¹

²³ Global Wind Energy Council, India Wind Energy Market Outlook 2022-2026, 2022

²⁴ Global Wind Energy Council, India Wind Energy Market Outlook 2022-2026, 2022

²⁵ Global Wind Energy Council, Accelerating Onshore Wind Capacity Addition in India to Achieve the 2030 Target, 2022

²⁶ Ministry of Petroleum and Natural Gas, Government has set target to increase share of gas in energy mix up to 15 per cent by 2030 : Union Minister of Petroleum & Natural Gas and Housing & Urban Affairs Shri Hardeep Singh Puri, 2023; Centre for Science and Environment, COMPRESSED BIOGAS (CBG) : Potential, Technology, Policy, Operations and Economics, 2023

²⁷ Ministry of Power, Power Sector at a Glance, 2023; Ministry of New and Renewable Energy, Renewable Energy in India, 2022

²⁸ ORF, The Implications of India's Revised Roadmap for Biofuels: A Lifecycle Perspective, 2022

²⁹ Ministry of Petroleum and Natural Gas, Biodiesel, 2022

³⁰ IEA Bioenergy, Implementation of bioenergy in India, 2021

³¹ Ministry of Petroleum and Natural Gas, National Policy on Biofuels-2018 Amendment, 2022

Green Hydrogen Energy

Green hydrogen (GH₂) energy is presently nascent in India. Green hydrogen is produced through an electrolysis process using electricity generated from renewable energy sources like solar, wind, or biomass. Currently, India annually consumes about 6 million metric tonnes (MMT) of hydrogen.³² Most of this is grey, meaning it is produced using fossil fuels such as natural gas and naphtha through a steam methane reforming process.

Figure: Key green hydrogen energy statistics



Source: NITI Aayog, Harnessing Green Hydrogen, 2022; Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023; Ministry of New and Renewable Energy, National Hydrogen Mission – An Update, 2023

India aims to be a global hub for green hydrogen production, usage, and export:

To achieve this, India plans to capitalise on its abundant renewable energy sources, implement favourable policies, make strategic investments, and foster international partnerships. The total financial layout for the mission between 2023-2030 is INR 19,744 crores, of which 87% is allocated for the Strategic Interventions for Green Hydrogen Transition (SIGHT) program, 7% for pilot projects, remaining 6% for research and development (R&D) and other miscellaneous components of the mission.³³

The industrial sector is the largest consumer of hydrogen in India:

As per a NITI Aayog report dated 2022, India's current hydrogen consumption stands at 6 million tonnes.³⁴ As mentioned above, all of this is grey hydrogen. Majority of the hydrogen produced is used for industrial purposes like oil refining, which accounts for 33% of total use, and ammonia production which is at 27%, as per 2020 figures. A smaller share of hydrogen use is attributed to methanol production which stands at 11%, followed steel production at 3%.³⁵

The green hydrogen sector can generate 6 lakh jobs by 2030:

Across the green hydrogen value chain, the supply of renewable energy such as solar and wind to produce green hydrogen is expected to create over 66% of new jobs. Plant construction and green hydrogen production will account for the second largest share of jobs, at 22% collectively.³⁶ Electrolyser manufacturing and distribution segment will account for 4% each, and 2% will be attributed to green hydrogen storage.³⁷

Atomic Energy

Atomic energy involves harnessing and utilising energy derived from atomic or nuclear reactions. Nuclear energy presently constitutes 1.5% of India's energy mix.³⁸ As per IESS estimates, this is expected to increase to 3.6% in 2047.³⁹ Presently, there are 22 nuclear reactors in operation, and 21 nuclear reactors are at various stages of development, with nine expected to be commissioned by 2024-25.⁴⁰

³² NITI Aayog, Harnessing Green Hydrogen, 2022; Department of Science and Technology, India Country Status Report on Hydrogen and Fuel Cells, 2020

³³ Ministry of New and Renewable Energy, Cabinet approves National Green Hydrogen Mission, 2023

³⁴ Note: Given that India's current production of hydrogen is 5 MMT, this figure assumes that India imports about 1 million tonnes of hydrogen

³⁵ NITI Aayog, Harnessing Green Hydrogen, 2022; Department of Science and Technology, India Country Status Report on Hydrogen and Fuel Cells, 2020

³⁶ Skill Council for Green Jobs, Gearing up the Indian workforce for a Green Economy, 2023

³⁷ Skill Council for Green Jobs, Gearing up the Indian workforce for a Green Economy, 2023

³⁸ NITI Aayog, India Energy Security Scenario, accessed December 2023

³⁹ NITI Aayog, India Energy Security Scenario, accessed December 2023

⁴⁰ Department of Atomic Energy, Proposals for New Atomic Power Plants, 2019
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Figure: Key atomic energy statistics



Source: NITI Aayog, India Energy Security Scenario (as under Least Effort Scenario – All Level 1), accessed December 2023; Department of Atomic Energy, Proposals for New Atomic Power Plants, 2019

Nuclear energy has a wide range of applications in India:

Nuclear power prevents the release of 41 million tonnes of CO₂ emissions annually in India when compared to the emissions that would result from generating equivalent electricity using coal-based thermal power plants.⁴¹ Along with energy generation, atomic energy in India is being used to treat cancer, sterilise medical products, and increase the shelf life of agricultural products like onions and mangoes.

Strict policy has led to higher safety standards:

India has implemented strict regulations and safety standards to ensure the safe functioning of nuclear power plants. The Atomic Energy Regulatory Board (AERB) oversees and enforces these measures. As a result, India's nuclear programme is one of the safest in the world with no single serious security incident in the past five decades.⁴²

India has commenced core loading of its first indigenous Fast Breeder Reactor (FBR) and should further explore Small Modular Reactors (SMRs):

Upon completion of core loading, India can subsequently generate power from its indigenously developed FBR. Once commissioned, India will be the second country after Russia to have a commercial operating FBR.⁴³ In addition to FBR, SMRs are a promising technology for industrial decarbonization, especially where a reliable and continuous power supply is essential. With capacities of up to 300 MW, SMRs provide design flexibility and enhanced safety compared to large nuclear plants. They also offer significant cost savings and shorter construction times since they can be factory-built, unlike conventional nuclear reactors that are constructed on-site. Exploring the participation of the private sector and startups in the development of this critical technology within India is essential.⁴⁴

Hydropower

Hydropower is one of India's biggest renewable energy sources. It involves energy generation via small hydropower plants, with a capacity of less than 25 MW, and large hydropower plants exceeding 25 MW in capacity.

Figure: Key hydropower statistics



Source: NITI Aayog, India Energy Security Scenario (as under Least Effort Scenario – All Level 1), accessed December 2023; Ministry of Power, FAQs on Hydropower, accessed December 2023

⁴¹ Department of Atomic Energy, Union Minister Dr. Jitendra Singh says, presently (2020-21) nuclear power is saving 41 million tonnes of CO₂ emissions annually, compared to the emissions that would be generated by equivalent electricity generation from coal based thermal power plants, 2023

⁴² Ministry of External Affairs, Nuclear Security in India, Accessed June 2023

⁴³ Prime Minister's Office, PM witnesses the historic "Commencement of Core Loading" at India's first indigenous Fast Breeder Reactor (500 MWe) at Kalpakkam, Tamil Nadu, 2024

⁴⁴ Department of Atomic Energy, Union Minister Dr. Jitendra Singh says, India taking steps for development of Small Modular Reactors (SMR), with up to 300 MW capacity to fulfill its commitment to Clean Energy transition, 2022
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India has expanded its hydropower capacity over the years:

The capacity of large hydropower increased from 39 GW in 2012 to 47 GW in 2023, while small hydropower rose from 1 GW in 2012 to 5 GW in 2023. Although the contribution of small hydropower remains relatively modest compared to other sources, it plays a crucial role in providing decentralized power to rural and remote areas, particularly in the North-eastern states.⁴⁵

Government policies and initiatives have supported hydropower development in India, attracting private sector participation:

Policies such as the Small Hydro Power Policy and the New Hydroelectric Policy have provided a supportive framework for developing and expanding hydropower projects in the country. Additionally, initiatives like the Hydro Purchase Obligation and the Renewable Purchase Obligation have stimulated market demand and encouraged investment in hydropower generation. The participation of private sector entities through joint ventures and public-private partnerships has brought in expertise, technology, and capital to accelerate the growth of the hydropower sector in India.

Emerging Technologies

Emerging technologies in the energy sector are opening new possibilities for sustainable and clean energy sources. While there is a wide range of emerging and early-stage technologies at various stages of development including synthetic hydrocarbon fuels, CCUS, advanced ultra super-critical power plants, coal gasification, Li-Air batteries, multivalent ion batteries, fusion nuclear, potassium-ion batteries, and biomass with CCUS, the Ministry of New and Renewable Energy (MNRE) focuses its programs on four new technologies: ocean energy, geothermal energy, hydrogen energy and energy storage technologies.

Figure: Key emerging technologies statistics



Source: Parliamentary Standing Committee on Energy (2020-21), Tidal Power Development in India, 2021; PIB, Power Minister Shri RK Singh approves proposal to declare ocean energy as Renewable Energy, 2019

India has significant potential for ocean, wave, and geothermal energy:

According to a study by IIT Madras, India has a theoretical potential of 12,500 MW for tidal energy and 41,000 MW for wave energy. The Geological Survey of India (GSI) has identified approximately 300 geothermal hot springs in India, with a potential of 10,000 MW. Together, these three energy sources have the potential equivalent to 15% of India's current installed power generation capacity.⁴⁶

India is investing in creating an enabling ecosystem for ocean, wave, and geothermal energy:

The Ministry of Earth Sciences has established the National Institute of Ocean Technology (NIOT) to focus on research and development in ocean-related technologies, including ocean energy. The government has also initiated pilot projects, such as the 65 KW OTEC (Ocean Thermal Energy Conversion) plant in Kavaratti, Lakshadweep, to explore the feasibility of harnessing ocean energy. The government has introduced measures in the geothermal sector that encompass opportunities for foreign direct investment, support in document processing, and incentives to encourage the exploration and development of geothermal resources.⁴⁷ It is also drafting a new Science, Technology, and Innovation Policy, which will be its first since 2013. Among the stated aims are to double the number of researchers and the level of public and private expenditure on R&D every five years, with energy as a priority topic.⁴⁸

⁴⁵ Comptroller and Auditor General of India, Union Civil Performance Renewable Energy Report – Chapter 5, 2015; Ministry of New and Renewable Energy, Annual Report, 2021-22; Ministry of Power, Growth of Electricity Sector in India From 1947-2020, 2020

⁴⁶ Ministry of New and Renewable Energy, Geo-Thermal and Ocean Energy Technologies, 2019

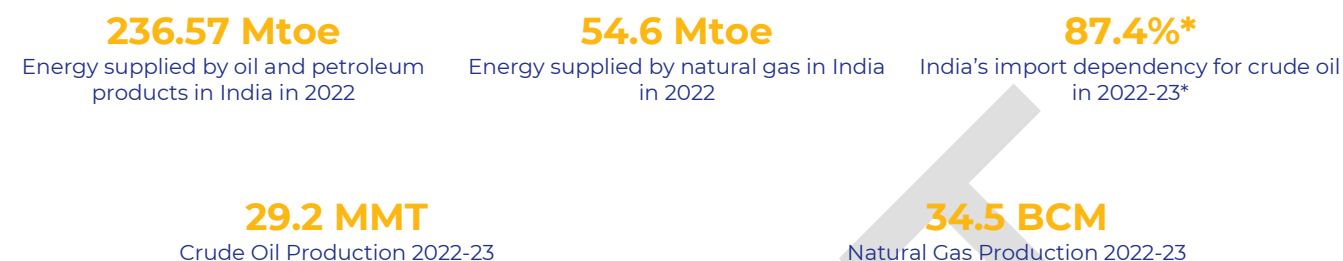
⁴⁷ Stanford University, Geothermal Energy in India: Poised For Development, 2018

⁴⁸ International Energy Agency, India Energy Outlook, 2021
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Oil and Natural gas

In India, oil and natural gas are vital in meeting the country's energy needs and driving economic growth. The oil sector has witnessed significant progress, with India being one of the largest consumers and importers of crude oil globally.

Figure: Key oil and natural gas statistics



Source: NITI Aayog, India Energy Security Scenario (as under Least Effort Scenario – All Level 1), accessed December 2023; Ministry of Petroleum and Natural Gas, India Hydrocarbon Outlook, 2023; Ministry of Petroleum and Natural Gas, Snapshot of India's Oil and Gas Data - Monthly Ready Reckoner – November, 2023
Notes: * On POL consumption basis; MTOE – million tonnes of oil equivalent; MMT – million metric ton; BCM – billion cubic metres

Oil and natural gas are expected to remain critical to our future needs.

Oil and petroleum products will contribute about 25% (724.46 mtoe) of India's total energy supply by 2047, and natural gas will contribute ~6% (176.92 mtoe).⁴⁹ The main drivers of demand for petroleum products are high-speed diesel oil (38%), petrol (15%) and LPG (14%). Additionally, natural gas use by the industrial sector and in buildings has further increased demand.⁵⁰

Domestic production of crude oil and natural gas has steadily declined over the past decade while imports and oil refinery capacity have risen:

India's crude oil reserves have declined marginally from 2014-15 and stand at about 600 million metric tonnes (MMT) as of 2021-21.⁵¹ India's oil refinery capacity has increased by 18% over the last decade, making it the world's fourth-largest crude oil refining hub as of 2023.⁵² India is unable to meet its oil demand through domestic production and imports most of its crude oil (87%), mainly from Iraq, Saudi Arabia, and the UAE.⁵³

Coal

Coal plays a significant role in India's energy landscape, and India is among the largest producers and consumers of coal globally. Coal is the dominant energy source in India, contributing the highest share to the country's energy mix.

Figure: Key coal statistics



Source: NITI Aayog, India Energy Security Scenario (as under Least Effort Scenario – All Level 1), accessed December 2023

⁴⁹ NITI Aayog, India Energy Security Scenarios 2047, Accessed September 2023

⁵⁰ Ministry of Statistics and Programme Implementation, Energy Statistics India, 2023; International Energy Agency, India Energy Outlook, 2021

⁵¹ Ministry of Petroleum and Natural Gas, Indian Petroleum & Natural Gas Statistics 2020-21, 2022

⁵² Ministry of Statistics and Programme Implementation, Energy – Statistical Year Book India, 2015; Ministry of Statistics and Programme Implementation, Energy Statistics India, 2023; Ministry of Petroleum and Natural Gas, History and Evolution, Accessed June 2023

⁵³ Petroleum Planning and Analysis Cell, Snapshot of India's Oil & Gas data, 2023; Ministry of Mines, Indian Minerals Yearbook 2020, 2022

Power generation and industrial processes drive demand for coal:

India ranks as the world's second-largest consumer of coal, just behind China. From 2012-13 to 2021-22, India witnessed an increase in coal consumption from 713 million tonnes to 1028 million tonnes. The major coal-consuming sectors include the power generation sector, accounting for the highest 69%, followed by steel & washery industries at 6%, and other industrial processes account for the remaining consumption.⁵⁴

Coal will continue to play a significant role in meeting India's short-term energy security requirements:

As per the India Energy Security Scenarios, coal's contribution to India's total energy supply will increase to 579.16 mtoe by 2027, and will further rise to 1536.24 mtoe by 2047.⁵⁵

Electricity

Over the past decade, India has made significant progress in expanding its electricity infrastructure and increasing access to reliable power supply. India has achieved nearly 100% household electrification, strengthening the distribution network and electricity access throughout the country.

Figure: Key electricity statistics



Source: Central Electricity Authority, General Review Report, 2023; NITI Aayog, India Energy Security Scenario (as under Least Effort Scenario – All Level 1), accessed December 2023; Ministry of Power, Saubhagya Dashboard, accessed July 2023; Ministry of Information & Broadcasting, India's Power Sector Transformation: A Journey Towards Sustainable Energy and Universal Access, 2023

Electricity consumption has experienced remarkable growth:

From 2012 to 2022, per capita electricity consumption increased from 884 kWh to 1255 kWh, and total electricity demand increased from 8,24,301 GWh to 12,96,300 GWh, at a CAGR of 5%, making India the third-largest consumer of electricity in the world.⁵⁶ The electricity consumption by the industry sector was highest at 41%, followed by the domestic and commercial sector at 34%, agriculture at about 18%, and transport and other applications consuming the rest.⁵⁷

Electricity is primarily sourced from coal, followed by renewables:

As of July 2023, thermal power (coal, lignite, diesel, and gas) comprised 56% of the total installed capacity⁵⁸, whereas renewable sources including solar, wind, biomass, small hydro, large hydro, and urban & industrial waste power account for 42%, and nuclear power accounts for 2% of total installed capacity.⁵⁹

Integrated Energy Assessment

Integrated energy assessment (IEA) provides data and insights to guide decisions related to energy sources, production and trading, and environmental regulations. It facilitates holistic planning, target

⁵⁴ Ministry of Statistics and Programme Implementation, Energy Statistics India, 2023

⁵⁵ NITI Aayog, India Energy Security Scenarios 2047, Accessed September 2023

⁵⁶ Ministry of Statistics and Programme Implementation, Energy Statistics India, 2023; Central Electricity Authority, Growth of Electricity Sector in India from 1947-2020, 2020; Ministry of Power, Ministry of Power implements significant schemes to increase energy efficiency, 2022

⁵⁷ Ministry of Statistics and Programme Implementation, Energy Statistics India, 2023

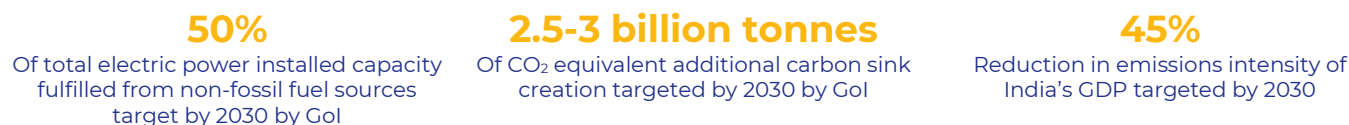
⁵⁸ CEA, All India Installed Capacity (In MW) of Power Stations, 2023

⁵⁹ CEA, All India Installed Capacity (In MW) of Power Stations, 2023

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setting for the short and medium term, and policy design to strengthen action at central and state levels across various ministries, departments, or agencies.

Figure: Key IEA statistics



Source: Ministry of Environment, Forest and Climate Change, India's Long-Term Low-Carbon Development Strategy, 2022; Cabinet, Cabinet approves India's Updated Nationally Determined Contribution to be communicated to the United Nations Framework Convention on Climate Change, 2022

The Prime Minister's Office (PMO) and NITI Aayog drive policymaking for integrated energy assessment:

Prime Minister's Office and the NITI Aayog, in collaboration with various central and state ministries and departments, work collaboratively to set a roadmap towards achieving the shared goals of energy security, improved access, affordability, efficiency, resource, and environmental sustainability. These entities also engage in activities like cross-sectoral ministerial coordination, drafting integrated energy policies, climate and energy security modeling, and coordinating efforts to expand energy statistics.

India has established multiple time-bound targets on energy transition, security, and access:

According to the revised Nationally Determined Contributions (NDC), India has pledged to decrease the emissions intensity of its GDP by 45% by 2030 compared to the 2005 level.⁶⁰ By 2030, India aims to fulfil 50% of its total electric power installed capacity from non-fossil fuel sources. India has also set the target to achieve net-zero emissions by the 2070.⁶¹

Energy Storage

Energy storage optimises energy systems by storing energy for later use, serving various applications like grid-level balancing for renewables, behind-the-meter (BTM) power backup inverters, and supporting the EV sector. The importance of energy storage technologies has grown significantly in India driven primarily by the increasing focus and ambition of integrating renewable energy rapidly into our energy mix.

Figure: Key energy storage statistics



Source: Ministry of Power, National Electricity Plan, 2023; Ministry of New and Renewable Energy, 64.54 billion units electricity produced from Wind Energy during April, 2022-January, 2023 – Union Power & NRE Minister, 2023; India Smart Grid Forum, Energy Storage System Roadmap for India: 2019-2032, 2019

India's energy storage demand is growing sharply, driven by renewables and EVs:

As of 2023, the overall energy storage capacity developed in the country is 4745.60 MW from Hydro Pumped Storage Projects (PSPs) and 39.12 MWh from Battery Energy Storage System (BESS). Energy storage in India witnessed a 23 GWh demand in 2018, 56% of which came from the power backup inverter segment. Furthermore, the EV industry accounted for more than 5 GWh of batteries in 2018.⁶²

⁶⁰ Cabinet, Cabinet approves India's Updated Nationally Determined Contribution to be communicated to the United Nations Framework Convention on Climate Change, 2022

⁶¹ Ministry of Environment, Forests and Climate Change, Net zero emissions target, 2023

⁶² India Smart Grid Forum, Energy Storage System Roadmap for India: 2019-2032, 2019; India Energy Storage Alliance, India Stationary Behind-the-meter (BTM) Energy Storage & Railway Battery Market Overview 2021-30, 2022; Ministry of Power, National Electricity Plan (Draft), 2022

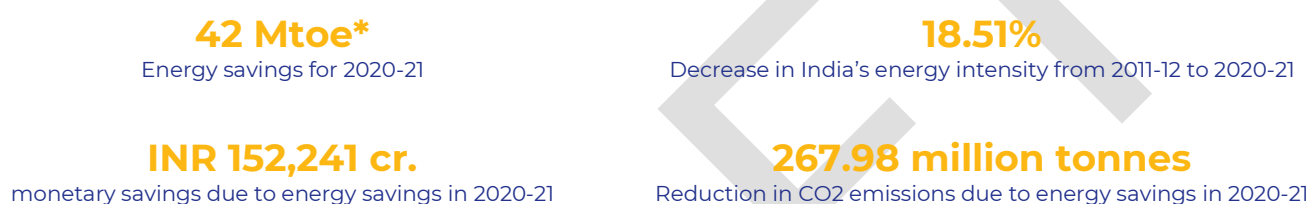
India's energy storage market is expected to expand rapidly in the near future:

During 2019-25, the combined potential for energy storage in grid side and behind-the-meter applications is estimated to be about 190 GWh.⁶³ Most of this large-scale energy storage will be driven by RE integration, Fast Response Ancillary Service (FRAS) market and T&D deferral. Separately, battery consumption by the EV industry is expected to be over 36 GWh by 2025.⁶⁴ Among the several factors contributing to this growth is the rising EV industry, increasing RE share necessitating stationary storage for grid stability, and BTM use cases like consumer and industrial electronics.⁶⁵

Energy Efficiency

Energy efficiency involves consuming less energy to achieve the same output while minimising waste. It encompasses various activities such as building energy efficiency, industrial energy management, appliance and equipment standards, transportation efficiency, process efficiency, agricultural efficiency, and energy recovery. It is vital for emissions reduction, and cost savings, and plays a pivotal role in India's sustainable development and energy security.

Figure: Key energy efficiency statistics



Source: Bureau of Energy Efficiency, Impact Assessment of Energy Efficiency for Year 2020-21, 2022; Ministry of Power, Energy Efficiency, Accessed July 2023
Note: Mtoe refer to million tonnes of oil equivalent

Government schemes are driving energy efficiency:

The Perform, Achieve, and Trade (PAT) scheme and Market Transformation for Energy Efficiency (MTEE) are two flagship programmes under the National Mission for Enhanced Energy Efficiency (NMEEE). The PAT scheme played a significant role in driving efficiency improvements in the industrial sector and led to energy savings of more than approximately 16 MTOE till 2022. Similarly, the Bachat Lamp Yojana of MTEE led to energy savings of about 4 billion units/year.⁶⁶

Energy efficiency policies have shown positive results:

Energy efficiency initiatives have resulted in a substantial energy savings of 42.0 million tonnes of oil equivalent (Mtoe), i.e., 4.73% of the country's total primary energy supply for 2020-21.⁶⁷ Additionally, from the years 2011-12 to 2020-21, India's energy intensity decreased from 0.27 megajoule (MJ) per INR to 0.22 MJ per INR, marking a significant decrease of 18.51%.⁶⁸

Negative Emission Technologies

Negative emission technologies (NETs) refer to innovative approaches and technologies that actively remove greenhouse gases – usually carbon dioxide – from the atmosphere. These technologies capture and store CO₂ or convert it into usable products via afforestation and reforestation, soil carbon sequestration, biochar, bioenergy with carbon capture and storage (BECCS) and direct air capture and carbon capture, utilisation, and storage (CCUS).

⁶³ India Smart Grid Forum, Energy Storage System Roadmap for India: 2019-2032, 2019

⁶⁴ India Smart Grid Forum, Energy Storage System Roadmap for India: 2019-2032, 2019

⁶⁵ NITI Aayog and Green Growth Equity Fund Technical Cooperation Facility, Advanced Chemistry Cell Battery Reuse and Recycling Market in India, 2022

⁶⁶ Ministry of Power, Status of Implementation of National Mission for Enhanced Energy Efficiency (NMEEE), 2022

⁶⁷ Bureau of Energy Efficiency, Impact Assessment of Energy Efficiency for Year 2020-21, 2022

⁶⁸ Bureau of Energy Efficiency, Impact Assessment of Energy Efficiency for Year 2020-21, 2022
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Figure: Key NET statistics

30%

Of all capturable emissions (i.e. 2400 mtpa) aimed to be capture by CCUS technologies by 2050 by GoI

1 billion tons

Aimed reduction in CO₂ emissions by 2030

Source: NITI Aayog, CCUS Policy Framework and its Deployment Mechanism in India, 2021; Ministry of Science and Technology, India is committed to achieve the Net Zero emissions target by 2070, 2023

India has recognised that CCUS will play a substantial role in its decarbonisation efforts:

India has set ambitious targets to combat climate change and reduce carbon dioxide (CO₂) emissions and is actively pursuing various strategies to decarbonise its economy. Recognising the significant role that Carbon Capture, Utilization, and Storage (CCUS) can play in this decarbonisation effort, India is focusing on developing and deploying CCUS technologies. India is actively exploring the potential for adopting CCUS in sectors like power generation from fossil fuels and hard-to-abate industry sectors such as steel.⁶⁹

⁶⁹ M. N. Dastur & Company and NITI Aayog, Carbon Capture, Utilization and Storage (CCUS), 2022, Ministry of Petroleum and Natural Gas, Carbon Capture Utilization & Storage, accessed June 2023
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Opportunities and challenges

Opportunities

Reliable access to affordable energy for all Indians	Provide electricity access in underserved regions: In remote regions with geographical and infrastructural challenges for centralized power grids, adopting decentralized energy solutions like off-grid solar, mini-grids, and small hydro projects can provide reliable electricity.⁷⁰ MNRE's Promotion of DRE (Decentralized Renewable Energy) Livelihood Applications framework aims to create an ecosystem for DRE adoption, supporting sustainable livelihoods in India.⁷¹ Reduce overall energy consumption: As per Ministry of Power, in 2020-21, India consumed 581.6 Mtoe of energy and saved 42 Mtoe due to energy efficiency measures, which is about 7% of the total energy consumed. The percentage saving in total energy consumption due to energy efficiency measures has consistently increased from <0.5% in 2011 to 7% in 2021.⁷² Improve electricity reliability: With vast potential in solar, wind, and hydro sources, India can sustainably meet rising electricity demands. Investing in modern grid infrastructure, smart grid solutions, and energy storage enhances grid stability, ensuring uninterrupted power, even in remote areas. The latest time-of-day tariff rules by the Government of India enable shifting elastic electricity demand to solar hours, improving reliability.⁷³ Further, since agricultural demand accounts for around 16% of the total electricity demand in 2022, a figure projected to remain at 10% in 2047, solarization in this sector has the potential to enhance both access to and the reliability of electricity supply for farmers.⁷⁴
Energy security and independence	Accelerate India's energy security journey: India's abundant natural energy resources offer a significant opportunity for long-term energy security. With an annual solar energy potential of around 5,000 trillion kWh, solar emerges as a highly secure energy source.⁷⁵ The country's surplus biomass, totaling 230 MMT annually, could generate approximately 28 GW of biopower. Leveraging the presence of 550 sugar mills, an additional 14 GW of power through bagasse-based cogeneration is feasible.⁷⁶ Further, hydropower plants provide rapid and cost-effective startup/shutdown, offering essential flexibility for addressing electricity demand variations.⁷⁷ While requiring substantial initial investments, hydropower plants boast extended and cost-effective lifespans, as exemplified by the Bhakra Nangal plant with operating costs as low as INR 0.10 per unit.⁷⁸ Reduce import dependency: According to projections by the IEA, if India successfully implements its energy efficiency policies and measures, the value of imports for oil, coal, and natural gas

⁷⁰ Ministry of New & Renewable Energy (MNRE), Evaluation of Small Hydro Power (SHP) Programme of MNRE, 2017

⁷¹ Ministry of New & Renewable Energy, Annual Report, 2021-22; Ministry of New & Renewable Energy, Framework for Promotion of Decentralised Renewable Energy Livelihood applications, 2022

⁷² Ministry of Power, National Electricity Plan, 2023

⁷³ Ministry of Power, Introduction of Time of Day Tariff, 2023

⁷⁴ NITI Aayog, India Energy Security Scenario, accessed on December 2023

⁷⁵ Ministry of New & Renewable Energy, Solar Energy, accessed July 2023

⁷⁶ Ministry of New and Renewable Energy, Bio Energy, 2022

⁷⁷ World Bank, India Hydropower Development, 2012

⁷⁸ World Bank, India Hydropower Development, 2012

	could be reduced by 18% in 2040, helping it avoid help us avoid tonnes of potential emissions and contribute to a sustainable energy future.⁷⁹ Take a globally leading role in the safe and sustainable development of nuclear energy: India, holding approximately 25% of the world's thorium reserves, is a pivotal player in atomic energy production. As the sole developing nation to successfully develop and deploy indigenous atomic reactors, India is actively addressing the global gap in commercial thorium reactor programs through significant investments in nuclear technology R&D.⁸⁰ This strategic effort positions India to emerge as a leading force in scaling viable solutions for global adoption. Continued investments in atomic energy, including the establishment of new nuclear reactors and incentives for domestic manufacturing, can elevate India to a prominent global leader in nuclear power production.⁸¹
Greater energy sustainability	Drive sustainable operations in the industrial and mobility sectors: Green hydrogen can directly replace fossil fuel-based resources like fuel or industrial feedstock. It can substitute grey hydrogen in industrial use cases like petroleum refining and fertiliser production. Further, it can be used in long-haul vehicles and marine vessels, replacing unsustainable fuels like diesel.⁸² Demonstrate global environmental stewardship in the transition from coal: India is advancing numerous clean coal initiatives, including Clean Coal Research and Development (CCORD) to enhance efficiency in coal-related processes and reclamation of mined-out lands.⁸³ The establishment of National Centres for Clean Coal Research and Development focuses on advancing research for efficiency and cost reduction. Additionally, the Advanced Ultra Super Critical plants aim to boost the efficiency of existing thermal plants.⁸⁴ India is also actively developing CCUS technologies for widespread adoption in heavy industries like steel, coal gasification, and thermal power plants.⁸⁵ Global leadership in CCUS technology: Given the global nascent state of CCUS technologies, India has a unique opportunity to become a global innovation hub for low-cost carbon capture methods. With access to vast onshore and offshore sedimentary basins for long-term carbon storage, strategic investments in R&D, business incubators, and collaborations with universities and foreign businesses can position India in global NET markets, offering long-term revenue opportunities for the CCUS industry.⁸⁶ While the Ministry of Petroleum and Natural Gas (MoPNG) emphasizes the role of carbon markets in industrial adoption, India's carbon markets are still nascent and need cost-effective development for widespread adoption.⁸⁷
Energy-led higher	Support the growth of other sectors like green hydrogen and electric mobility: The growing focus on green hydrogen which will help India become a low-carbon economy, translates to the importance of renewables like solar energy used in its

⁷⁹ IEA, Energy Efficiency Outlook for India – Sizing up the opportunity, 2016

⁸⁰ Press Information Bureau, Thorium-Based Nuclear Reactors, 2019

⁸¹ Department of Atomic Energy, Union Minister Dr Jitendra Singh says, nearly 9% share of electricity likely to be contributed from India's nuclear sources by the year 2047; Department of Atomic Energy,

⁸² Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023

⁸³ Ministry of Coal, Greening Initiatives of India's Coal Sector, 2022; Ministry of Science and Technology, Clean Coal Research Initiative (CCRI), Accessed July 2023; Ministry of Science and Technology, Clean Coal Research Initiative (CCRI), Accessed July 2023

⁸⁴ Ministry of Heavy Industries, Development of Advance Ultra Supercritical Technology, 2023

⁸⁵ NITI Aayog, CCUS Policy Framework and its Deployment Mechanism in India, 2021

⁸⁶ Ministry of Petroleum, Oil and Natural Gas, Draft 2030 Roadmap for Carbon Capture Utilization and Storage (CCUS) for Upstream E&P Companies, 2022

⁸⁷ Ministry of Petroleum, Oil and Natural Gas, Draft 2030 Roadmap for Carbon Capture Utilization and Storage (CCUS) for Upstream E&P Companies, 2022

production. The decline in costs of technologies like solar PV and electrolyzers is considered one of the main driving factors for the achievement of cost-competitiveness by green hydrogen.⁸⁸ Similarly, the powering of EVs by renewables and growth in battery storage systems will help India achieve green mobility in its true sense. Solar rooftop PV is an attractive technology for cost-effective and scalable deployment in India's EV sector.⁸⁹

Establish India as a first mover and future leader in global GH2 markets:

The global demand for green hydrogen and its derivatives is expected to rise in line with the net zero revolution. India has a rich opportunity to become a leader in international trade and cater to this increasing demand by leveraging the country's abundant resources.⁹⁰

Participate and lead in global solar energy innovation:

Advancements in manufacturing processes in recent years have increased solar efficiency in India, bringing down the levelised cost of electricity.⁹¹ Further, India is already taking steps to capture the global stage in the solar energy system. Through the International Solar Alliance (ISA), it fosters global collaboration to accelerate grid interconnectivity, mobilise investment, and support deployment through capacity-building.⁹² India must continue to engage global partners and emerge as a key contributor to drive and lead innovation in solar energy.

Huge potential for green jobs :

As per a CEEW report, India has the potential to create about 3 million jobs (short and long-term) by increasing its installed capacity of solar and wind to achieve the overarching goal of 500 GW.⁹³ Thus, India has a significant opportunity to facilitate the smooth transition of the existing fossil fuel workforce to renewable energy sectors.

Position India as a global wind component export hub:

India has already established itself as the world's second-largest hub for wind supply chain manufacturing, and further expansion in this sector will lead to additional job opportunities, bolster GDP growth, and foster technical advancements. This aligns with the government's "Make in India" initiative, promoting domestic manufacturing capabilities and reinforcing the country's position as a key player in the global wind energy market.⁹⁴

Boost domestic manufacturing of batteries:

Establishing a robust domestic battery manufacturing ecosystem, including a strong cell supply chain, is vital for India's competitiveness in mobility, grid energy storage, and consumer electronics. This move aims to decrease the escalating import costs of li-ion cells, guard against potential supply shocks, and foster job creation. Private sector investment opportunities exist through government partnerships, facilitating risk-sharing. Incentives for manufacturers are already in motion through the approved PLI scheme of 2021.⁹⁵

88 The Energy and Resources Institute, The Potential Role of Hydrogen in India: A pathway for scaling-up low carbon hydrogen across the economy, 2020

89 Center for Study of Science, Technology and Policy, Solar Energy-Based EV Charging: A Pilot and Techno-Economic Study, 2021

90 Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023

91 CEEW, Making India a Leader in Solar Manufacturing, 2022

92 ISA, ISA's Solar Banking Initiative, accessed June 2023

93 CEEW, India's Expanding Clean Energy Workforce, 2022

94 Global Wind Energy Council, India Wind Energy Market Outlook 2022-2026, 2022

95 Cabinet, Cabinet approves Production Linked Incentive scheme "National Programme on Advanced Chemistry Cell Battery Storage", 2021; NITI Aayog and RMI, Need for Advanced Chemistry Cell Energy Storage in India, 2022

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Challenges

Demand-side challenges	- Limited awareness and adoption of energy-efficient appliances persist in India, with only one in four households recognizing the Bureau of Energy Efficiency's star label, especially in rural areas, where awareness is at 19%.⁹⁶ - Public resistance, fueled by incidents like the Fukushima nuclear disaster, hinders the acceptance of new nuclear power plants, causing delays, as seen in the Kudankulam plant's commercial operation delayed by over six months due to local protests.⁹⁷ - Despite recent progress, more than 400 million individuals in India still rely on traditional cooking fuels due to infrastructural gaps, low connectivity, and the high cost of LPG.⁹⁸
Fossil-sources related challenges	- Lower quality of domestic coal because of lower energy, higher ash content, and varying moisture levels makes India dependent on imports to meet 90% of coking coal requirement in the steel sector.⁹⁹ - The transition to cleaner energy poses a threat to coal workforce livelihoods, especially for informal workers lacking protection under current labour laws, highlighting the need for a comprehensive Just Transition roadmap with policies for economic impact mitigation and diversification options.¹⁰⁰ - The decline in domestic crude oil and natural gas production due to depleting reserves and ageing fields has led to a significant rise in oil and gas imports, with crude oil import increasing from 82% to 86% of total demand and natural gas import rising from 30% to 47% over the period from 2012-13 to 2021-22.¹⁰¹ - India's oil and gas sector growth has increased hydrocarbon emissions, and while carbon abatement technologies like CCUS offer potential, challenges include the lack of natural CO₂ sources, high costs, and the need for substantial infrastructure, policy support, technology transfers, and financing from developed economies to achieve net-zero emissions targets.¹⁰²
Clean-energy sources related challenges	- Domestically manufactured solar modules in India are 33% more expensive than Chinese counterparts due to higher bill of materials costs (56% of the price difference), lower manufacturing scale efficiency, higher labor and overhead costs, and underutilization of existing manufacturing capacity at 50%.¹⁰³ - Land acquisition hurdles persist in both utility-scale solar and wind projects in India, marked by declining availability of suitable wasteland, lengthy approval processes for agricultural land use, and state restrictions on land acquisition, leading to delays, increased costs, and hindering investment decisions for renewable energy developers.¹⁰⁴ - Scarce state policy focus and varied ownership structures impede wind repowering projects in India, with only a handful of states having repowering elements.¹⁰⁵ - Limited and fragmented data on biomass availability hampers project planning and cost.¹⁰⁶

96 CEEW, Awareness and Adoption of Energy Efficiency in Indian Homes, 2020

97 International Atomic Energy Agency, Challenges in developing Nuclear Power Infrastructure, 2017; Department of Atomic Energy, Target for Atomic Energy, 2017

98 World bank, Access to clean fuels and technologies for cooking, 2021

99 PIB, Government of India taking steps to raise the level of availability of domestic coking coal for use in steel industries, 2023

100 iForest, Just transition of coal-based power plants in India, 2022

101 Ministry of Petroleum and Natural Gas, Indian Petroleum and Natural Gas Statistics, 2022; Ministry of Statistics and Programme Implementation, Energy Statistics India, 2023

102 Indian Council for Research on International Economic Relations (ICRIER), Exploring Carbon Capture, Utilisation and Storage in the Indian Context: ICRIER G20 Policy Series - Run Up to 2023, 2022

103 CEEW, Scaling Up Solar Manufacturing to Enhance India's Energy Security, 2020; JMK Research and IEEFA, Viability Assessment of New Domestic Solar Module Manufacturing Units, 2021

104 IEA, India Energy Outlook, 2021; Council on Energy, Environment and Water (CEEW), How India's Solar and Wind Policies have Enabled its Energy Transition, 2021

105 Mongabay, After little success in past, India revisits plan to repower its ageing wind turbines, 2022

106 United Nations Centre For Regional Development, Discussion paper on Potential of economic utilization of biomass waste in India - Implications towards SDGs, 2016

	• Biofuel expansion in India faces hurdles like limited domestic sources, uncertain biomass supply, and market barriers, including constraints in supply chain logistics, feedstock inadequacy, and import restrictions, hindering the increased utilization of biofuels.¹⁰⁷ • The current production cost of green hydrogen is over twice that of fossil fuel-derived hydrogen making it economically unviable in short term but decreasing renewable energy prices and potential economies of scale offer future prospect.¹⁰⁸ • Restrictions on private sector participation in nuclear power hinder its growth, with limited involvement currently confined to equipment manufacturing and support services.¹⁰⁹ • The high cost of power evacuation in remote locations, compounded by the absence of proper infrastructure and reliance on low-capacity rural networks, imposes financial burdens on small hydropower projects, hindering their development.¹¹⁰ • Inadequate grid infrastructure impedes large-scale renewable energy integration, causing grid instability and curtailment.¹¹¹ • The significant land requirement for solar energy, roughly 2% of India's total surface area, poses challenges to India's ecology and can lead to social conflicts.¹¹²
New technologies and innovation related challenges	• Domestic battery manufacturing face hurdles in India due to low reserves of essential components, limited technological knowledge, and dependence on imports, with li-ion batteries constituting a significant portion of EV costs and patent constraints hindering technology transfer.¹¹³ • High costs of NETs, particularly in early development stages, pose a barrier to widespread adoption, with variations in CO2 capture technologies contributing to additional cost complexities for different applications.¹¹⁴ • Assessing India's ocean energy potential faces difficulties due to complex coastal dynamics, geography, seasonal and regional wave patterns, and limitations in on-site measurements, hindering accurate evaluation and development.¹¹⁵ • High capital costs, ranging from INR 30-60 crore per MW for ocean energy and Rs 30 crore/MW for geothermal, impede the adoption of these emerging technologies, making them financially non-competitive compared to mature energy sources.¹¹⁶
Governance and planning challenges	• Limited coordination between institutions in India's energy sector hinders effective integrated energy assessment, posing difficulties in aligning diverse objectives and adapting to dynamic regulatory and economic conditions.¹¹⁷ • Inadequate regulatory attention to battery use and safety standards hinders the market competitiveness of storage solutions, limiting revenue generation and compromising safety in India's operating conditions.¹¹⁸ • Insufficient institutional capacity for energy efficiency (EE) implementation is evident in limited EC cells and unit offices, emphasizing the need for broader collaboration with State Designated Agencies (SDAs) and regular capacity-building programs for effective local implementation.¹¹⁹ • Insufficient policies and incentives impede widespread CCUS adoption, discouraging investment and hindering infrastructure development. A lack of regulatory mechanisms and an early-stage Indian Carbon Market add further constraints to CCUS implementation in India.¹²⁰

¹⁰⁷ International Energy Agency, Commercializing Conventional and Advanced Transport Biofuels from Biomass and Other Renewable Feedstocks, 2020

¹⁰⁸ NITI Aayog, Harnessing Green Hydrogen – Opportunities for Deep Decarbonisation in India, 2022

¹⁰⁹ Observer Research Foundation, India's targets for nuclear energy: Moving closer?, 2023

¹¹⁰ Ministry of New & Renewable Energy (MNRE), Evaluation of Small Hydro Power (SHP) Programme of MNRE, 2017

¹¹¹ International Energy Agency, Renewables Integration in India, 2021

¹¹² IEEFA, Renewable Energy and Land Use in India by Mid-Century, 2021

¹¹³ NITI Aayog and RMI, Need for Advanced Chemistry Cell Energy Storage in India, 2022; Ministry of Heavy Industries, Evaluation of Electric Vehicle (EV) Policy, 2023

¹¹⁴ NITI Aayog, CCUS Policy Framework and its Deployment Mechanism in India, 2021

¹¹⁵ Indian Institute of Technology, Bombay, Estimating Ocean Renewable Energies for Potential Harvesting, Accessed June 2023

¹¹⁶ Ministry of New and Renewable Energy, Geo-Thermal and Ocean Energy Technologies, 2019

¹¹⁷ International Energy Agency, Understanding Energy Challenges in India, 2012

¹¹⁸ IESA and WBCSD, Policy and Regulatory Framework to Accelerate Energy Storage Deployment in India, accessed June 2023

¹¹⁹ Ministry of Power, State Energy Efficiency Index, 2021-22

¹²⁰ Ministry of Power, Ministry of Power & Ministry of Environment, Forests & Climate Change to develop Carbon Credit Trading Scheme for Decarbonisation, 2023; Ministry of Petroleum and Natural Gas, Draft 2030 Roadmap for Carbon Capture Utilization and Storage (CCUS) for Upstream E&P Companies, accessed June 2023

Fiscal challenges	• Fossil fuels contribute significantly to government revenues in India, accounting for 20.8% of central government revenue and 8.3% of state government revenue in 2019. According to estimates by the Taskforce on Climate and Development and the IMF, as India transitions towards renewable energy sources, the revenue from fossil fuels is expected to decrease by 13-15 percentage points for the central government and by 5-6 percentage points for state governments.¹²¹ • A 2022 report by the Standing Committee on Energy highlighted a substantial investment gap in meeting India's renewable energy targets for 2030. While the required annual investment is between Rs 1.5-2 lakh crore, the actual annual investment has been only Rs 75,000 crore.¹²² • Given the lack of understanding of the risks and returns, only a few domestic financial institutions and banks provide financial assistance to the renewable energy sector.¹²³
Challenges faced by key energy intermediaries	• Energy Service Companies (ESCOs) face financing constraints, hindering their role in overcoming critical barriers in the energy efficiency market, as they struggle to secure finance due to a lack of specific guidelines on bank loans from the RBI.¹²⁴ • India's power sector faces persistent high AT&C losses at 20%, adversely impacting the financial health of DISCOMs. Despite efforts like the UDAY scheme, DISCOMs continue to grapple with recurring losses, hindering the required INR 33,60,594 crore investments for essential network improvements and renewable energy integration.¹²⁵

¹²¹ Task Force on Climate and Development, IMF, India's Energy and Fiscal Transition, 2022

¹²² Standing Committee on Energy, Financial Constraints in Renewable Energy Sector, 2022

¹²³ Standing Committee on Energy, Financial Constraints in Renewable Energy Sector, 2022

¹²⁴ Sarkar A. and Sundararajan S., Transforming India's energy efficiency market by unlocking the potential of private ESCOs, 2020; CEEW and National Resources Defense Council (NRDC), Greening India's Financial Market: Investigating Opportunities for a Green Bank in India, 2016; Small Industries Development Bank of India (SIDBI), REoI for Partial Risk Sharing Facility for Energy Efficiency Project, 2020

¹²⁵ PRS Legislative Research, Overview of the Power Sector, 2019; PRS Legislative Research, Overview of the Power Sector, 2019
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Institutional framework and policy landscape —

Key ministries and departments

The centre and states play equally important but differentiated roles in the energy ecosystem in India. While electricity is a concurrent subject governed jointly by the centre and the states, sectors like coal, oil, natural gas, and atomic energy fall under the sole jurisdiction of the central government. Though not the primary decision-makers, ministries such as heavy industries, agriculture, environment, and road transport, are critical stakeholders that wield substantial influence over energy policies and practices.

The central government has the primary agenda of designing policies, setting targets, and monitoring across all aspects of the energy sector. The Ministry of Power in India oversees the development of electrical energy. It houses regulatory organisations like the Bureau of Energy Efficiency, the Central Electricity Authority, and the Central Electricity Regulatory Commission. These entities work towards enhancing energy efficiency and ensuring a reliable and quality power supply to all consumers. Public Sector Undertakings like NTPC and Power Finance Corporation synergise efforts to ensure reliable, resilient, and sustainable power systems, fostering innovation, efficiency, and affordable financing for the Indian power sector.

The Ministry of New and Renewable Energy (MNRE) in India is dedicated to advancing the development and utilisation of new and renewable energy sources to supplement the country's energy requirements. Key sub-organisations within the ministry include national solar and wind energy institutes and the Indian Renewable Energy Development Agency. These entities primarily focus on accelerating the renewable energy sector's growth by developing solar and wind energy technologies and strengthening financing in this sector.

The Ministry of Petroleum and Natural Gas focuses on enhancing domestic petroleum resources' production and utilisation, addressing priorities like energy sustainability. The Petroleum Planning and Analysis Cell, Directorate General of Hydrocarbon, and Petroleum and Natural Gas Regulatory Board operate under the ministry's purview. Collectively, they are responsible for establishing robust data and policy analysis practices, effectively managing resources, and ensuring fair trade to develop vibrant energy markets. Public Sector Utilities (PSUs) like Oil and Natural Gas Corporation (ONGC), Hindustan Petroleum Corporation Limited (HPCL), and Bharat Petroleum Corporation Limited (BPCL) operate to ensure the development, production, and sale of petroleum resources. They also integrate the exploration, refining, and marketing of hydrocarbons while focusing on environmental responsibility for sustainable growth.

The Ministry of Coal ensures a reliable supply of coal to meet the energy requirements of different sectors of the economy in an environmentally friendly and cost-effective manner. Under the ministry, the Coal Controller's Organisation aims to ensure fairness in production and commercial transactions within India's coal sector. Coal India Limited aims to efficiently produce and market coal and coal products, focusing on safety, conservation, quality, and eco-friendliness.

Due to its strategic importance, the Department of Atomic Energy remains under the control of the Prime Minister's Office (PMO) and is responsible for managing the atomic resources of the country, including their usage to meet the energy and non-energy requirements of various sectors. The Atomic Energy Regulatory Board (AERB) aims to safeguard public health and the environment by overseeing the safe utilisation of ionising radiation and nuclear energy in India, mitigating any potential risks.

Several departments, agencies, and PSUs oversee and govern the energy sector at the state level. The State Electricity Regulatory Commissions (SERCs) oversee electricity generation, distribution, and transmission within states and union territories of India, ensuring compliance and facilitating inter-state

electricity transmission. In addition to this, State Transmission Utilities (STUs) are responsible for intra-State electricity transmission, including planning, coordination, and development of the intra-state transmission system in collaboration with Central Transmission Utilities, State Governments, and other stakeholders. DISCOMs (Distribution Companies) act as intermediaries between electricity generators and end consumers, ensuring efficient distribution of electricity through operating, maintaining, and managing distribution systems, facilitating energy purchase from generators, and supplying it to agricultural, industrial, commercial, and residential users. There are entities working towards developing specific energy sources, like State Nodal Agencies (SNAs) which focus on promoting renewable energy sources in their state and a transition to cleaner fuels.

Several central and state-level bodies work together to ensure coordinated action in the energy sector. Falling under the chairmanship of the Prime Minister, the NITI Aayog supports policy analysis and coordination across ministries and states to advance energy policy objectives. Additionally, a Forum of Regulators (FOR) serves as a platform for coordination among the various state regulatory agencies and the central regulator. Companies like Power System Operation Corporation (POSOCO) and National, Regional and State Dispatch Centres (NLDC, RLDC, SLDC) operate together to ensure grid security and balance.

Key ministries and departments



NITI Aayog

Conducts policy analysis of energy sectors related to power, coal, petroleum and natural gas, new and renewable energy, and atomic power; and undertakes steps to improve energy planning, and data management in India



Ministry of Power

Administers, formulates and implements power sector-related policies and provides R&D and technical assistance relating to hydroelectric and thermal power, transmission system networks and distribution systems in the States/UTs



Ministry of New and Renewable Energy (MNRE)

Develops renewable and clean energy technologies, systems, products, and services in line with global standards and deploys these for energy applications for sectors like power, transport and support country's objective of achieving energy security



Ministry of Petroleum and Natural Gas

Undertakes exploration and extraction of oil and natural gas, as well as the refining, distribution, and marketing of petroleum products, alongside handling activities related to the import, export, and conservation of these products



Ministry of Coal

Determines and implements policies and strategies related to the exploration and mining of coal and lignite reserves and carries out activities like power generation, distribution, and marketing through entities under its administrative control like CIL



Department of Atomic Energy

Carries out various activities to develop nuclear power technology like mineral exploration, operation of nuclear power plants and waste management, alongside spearheading R&D in advanced power-related technology and in non-power applications of atomic energy



Bureau of Energy Efficiency

Assists in developing policies and strategies to promote energy efficiency and conservation and utilises existing resources and infrastructure in performing functions under the Energy Conservation Act



Central Electricity Authority

Acts as a technical support base for power sector stakeholders and supports policy formulation, establishing technical standards, project monitoring and human resource skilling



Central Electricity Regulatory Commission

Improves regional transmission system operations, formulates efficient tariff mechanisms, promotes power market development, and facilitates technological advancements alongside environmental and safety compliance



National Institute of Solar Energy

Works towards research and development activities, conducts testing and certification of solar components, carries out training programs, and fosters the growth of solar products and applications



National Institute of Wind Energy

Engages in research and development related to wind energy, conduct wind resource assessment (both onshore and offshore), establishes standards, and carries out testing, certification, and skill development initiatives for the wind energy sector



Indian Renewable Energy Development Agency

Finances and supports self-sustainable investments in energy generation from renewable sources, and carry out promotion of energy efficiency and sustainable environmental technologies



Petroleum Planning and Analysis Cell

Adopts the latest technology for strengthening data systems in oil and gas sector, assists in oil and gas pricing and administration of schemes, and analyses domestic and international energy markets



Directorate General of Hydrocarbon

Implements policies and contracts concerning exploration of coal, oil, and natural gas, promote investments, conduct monitoring of exploration and production activities alongside the development of non-conventional hydrocarbon energy sources



Petroleum and Natural Gas Regulatory Board

Facilitates investment flow into infrastructure for efficient transportation and distribution of petroleum and natural gas and promotes fair trade practices to ensure enhanced competitiveness of Indian markets and customer satisfaction



Coal Controller's Organization

Collects and maintains coal production data of all private and public sector coal mines in the country



Atomic Energy Regulatory Board

Conducts review of nuclear and industrial aspects of entities involved in nuclear sector, drafts safety policies and codes, and ensures compliance with its regulations



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Conducts policy analysis of energy sectors related to power, coal, petroleum and natural gas, new and renewable energy, and atomic power; and undertakes steps to improve energy planning, and data management in India

Major policies and programs

This section highlights existing policies related to the water sector and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.



Mission Coking Coal:

This Ministry of Coal has launched Mission Coking Coal to boost domestic coking coal production by allocating new coking coal blocks to private players, revitalizing abandoned mines, adopting advanced technologies, improving quality parameters, and enhancing coking coal washeries. The initiative seeks to strengthen in-house capabilities, reduce coking coal imports, and support the goal of AtmaNirbhar Bharat by ensuring a steady supply of high-quality coking coal for the country.¹²⁶



Coal Gasification Mission:

The Ministry of Coal aims to achieve 100 million tonnes of coal gasification by 2030 under the National Coal Gasification Mission. To achieve this, it lays down several directives such as mapping the gasification potential of current coalfields, developing gasification technology suitable for different varieties of coal, developing marketing strategies for end products of the process, and providing quantifiable metrics to monitor implementation at various companies.¹²⁷



National Wind Solar Hybrid Policy, 2018¹²⁸

This policy provides a framework for the promotion of large grid-connected wind-solar PV hybrid systems for efficient utilisation of transmission infrastructure and land. It aims to reduce the variability in renewable power generation and achieving better grid stability



National Offshore Wind Energy Policy, 2015¹²⁹

The Policy aims to establish offshore wind power projects through:

- Promoting the deployment of offshore wind farms in the Exclusive Economic Zones (EEZ) of the country
- Encouraging spatial planning and management of maritime renewable energy resources in the exclusive economic zone of the country through suitable incentives

Enhancing the replenishment of groundwater sources and implementing sustainable water conservation practices

¹²⁶ Ministry of Coal, Mission Coking Coal - Recommendations of Inter- Ministerial Committee For Augmenting Coking Coal Production, 2021; Ministry of Coal, Strategic Initiatives by Coal Ministry increase Availability of Domestic Coking Coal for Steel Production, Reduce Imports, 2023

¹²⁷ Ministry of Coal, National Coal Gasification Mission, 2021

¹²⁸ Ministry of New and Renewable Energy, National Wind-Solar Hybrid Policy, 2018

¹²⁹ Ministry of New and Renewable Energy, Wind Policies, 2015

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(Amended) National Policy on Bio-Fuels, 2022¹³⁰

The Policy enables the availability of biofuels in the market by:

- Proposing a target of 20% blending of ethanol in petrol by Ethanol Supply Year (ESY) 2025-26
- Reducing the import of petroleum products by fostering domestic biofuel production

(Notified) National Bioenergy Programme, 2022¹³¹

The programme aims to increase bioenergy usage through:

- Supporting the installation of large biogas-, bio-CNG power- plants
- Promoting the use of pellets and briquettes power generation and non-bagasse-based power generation projects
- Setting up of family and medium size biogas installations in rural areas
- Implement Waste to Energy programme to support generation of BioCNG from urban, industrial, and agricultural waste

National Green Hydrogen Mission, 2023¹³²

The mission aims to position India as a global leader in the production, use and export of green hydrogen through the following key provisions:

- Ramp up domestic capacity to annually produce green hydrogen up to a minimum of 5 MMT by 2030 by robust R&D efforts, reforming regulations, and pilot programs
- Develop and scale production-related technology, and build the infrastructure required for storage and delivery to make green hydrogen accessible and affordable

Mission on Advanced and High-Impact Research (MAHIR), 2023¹³³

The mission, launched jointly by the Ministry of Power and the Ministry of New and Renewable Energy, strives to foster indigenous research, development, and demonstration of the latest and emerging technologies in the power sector. The mission aims to harness emerging technologies and advance them to the implementation stage, with the goal of utilizing them as the primary catalyst for India's future economic growth and establishing the country as a global manufacturing hub. The following technologies are prioritized for research: alternatives to lithium-ion storage batteries, modifying electric cookers and pans to suit Indian cooking methods, green hydrogen for mobility, carbon capture, geothermal energy, solid-state refrigeration, nano-technology for EV batteries, and indigenous cold-rolled grain-oriented steel technologies.¹³⁴

Scheme for Harnessing and Allocating Koyala Transparently in India (SHAKTI), 2017¹³⁵

The scheme aims to decrease reliance on imported coal, ensuring the consistent availability of coal to thermal plants through transparent and objective processes. It also aims to guarantee that the benefits of linked coal are transferred to end consumers.

¹³⁰ Ministry of Petroleum and Natural Gas, National Policy on Biofuels-2018 Amendment, 2022

¹³¹ Ministry of New and Renewable Energy, MNRE notifies National Bio Energy Programme, 2022

¹³² Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023

¹³³ Ministry of Power, Mission on Advanced and High-Impact Research (MAHIR) launched to leverage Emerging Technologies in Power Sector, 2023

¹³⁴ Ministry of Power, Mission on Advanced and High-Impact Research (MAHIR) launched to leverage Emerging Technologies in Power Sector, 2023

¹³⁵ Ministry of Coal, SHAKTI, 2017

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National Mission for Enhanced Energy Efficiency, 2011¹³⁶

This mission consists of four specific energy efficiency initiatives under its umbrella that are aimed at facilitating energy intensive industries (industries that consume large quantities of energy in production process) to reduce their energy footprint and catalyse investments in the energy efficiency sector.



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Ujwal DISCOM Assurance Yojana (UDAY), 2015¹³⁷:

The UDAY scheme is designed to achieve a financial turnaround of DISCOMs (Distribution Companies) in alignment with the government's goal of ensuring 24X7 affordable and accessible electricity for all. It aims to enhance operational efficiency, reduce power costs and interest expenses for DISCOMs, and enforce financial discipline through collaboration with state finances. States have the option to participate in the scheme by signing a tri-partite agreement with the Ministry of Power and their respective DISCOMs.¹³⁸



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Perform, Achieve, and Trade (PAT) Scheme¹³⁹:

The scheme is aimed at reducing Specific Energy Consumption (SEC) in energy-intensive industries. To achieve this, a market-based mechanism is employed allowing certification of excess energy savings which can be traded, thereby enhancing cost-effectiveness. Implementation of PAT cycle-II from 2016-19 resulted in total energy savings of about 14.08 million tonnes of oil equivalent (MTOE), translating into avoiding about 68 million tonnes of CO₂ emission.¹⁴⁰



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Pradhan Mantri Ujjwala Yojana (PMUY)¹⁴¹:

The scheme aims to release deposit-free LPG connections in the name of adult woman members of poor households in India. As on 01.07.2023, there are close to 9.5 crore PMUY beneficiaries, out of which about 8.4 crore have taken at least one refill during 2022-23. Under Ujjwala 2.0, a special provision has been made for migrant families who can use a self-declaration instead of address proof and ration card to apply for a PMUY connection.



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Hydrocarbon Exploration and Licensing Policy (HELP), 2016¹⁴²:

HELP seeks to increase domestic oil and natural gas production in India and reduce import dependency by enabling greater ease of doing business. It aims to provide greater freedom to industry operators for effective exploration of existing hydrocarbon resources. It also grants a uniform license to investors to explore and extract hydrocarbon resources. Under the HELP regime, several exploration activities related to unconventional hydrocarbons have been undertaken in the last five years.¹⁴³



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National Mission on Transformative Mobility and Battery Storage, 2019¹⁴⁴

The mission aims to promote environmentally friendly and sustainable mobility initiatives. This includes implementing a Phased Manufacturing

¹³⁶ Bureau of Energy Efficiency, National Mission for Enhanced Energy Efficiency, accessed June 2023

¹³⁷ Government of India, UDAY for financial turnaround of Power Distribution Companies, 2015

¹³⁸ Government of India, UDAY for financial turnaround of Power Distribution Companies, 2015

¹³⁹ Bureau of Energy Efficiency, Perform Achieve Trade, accessed June 2023

¹⁴⁰ Bureau of Energy Efficiency, Perform Achieve Trade, accessed July 2023

¹⁴¹ Ministry of Petroleum & Natural Gas, PRANDHAN MANTRI UJJWALA YOJNA (PMUY), 2023

¹⁴² Ministry of Petroleum and Natural Gas, Hydrocarbon Exploration and Licensing Policy (HELP) – A Win-Win approach, accessed June 2023

¹⁴³ Ministry of Petroleum and Natural Gas, Primary objective of the HELP is ease of doing business and operational freedom to Operators, 2022

¹⁴⁴ Cabinet, Mobility Solutions get a boost, 2019

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Programme (PMP) to help establish battery and cell manufacturing giga plants that are large-scale and export competitive. Furthermore, another PMP encourages localised production throughout the entire value chain of Electric Vehicles (EVs).

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Way forward

India has made substantial progress in advancing energy security, sustainability, affordability, and accessibility for all citizens. However, as highlighted in the above sections, much more progress is required to realize its ambitions of net-zero India by 2070 while meeting its commitments to equity and climate justice. Some key areas of intervention for energy policymakers going forward include the following:

 Accelerate renewables deployment and infrastructure	- Policymakers can enable the development of state-level plans that outline specific targets for the expansion of each type of renewable energy and sector-wise decarbonization efforts. This plan could include details on the expected capacity additions, timelines, and the regions where these developments will take place. - Policymakers could focus on enhancing grid flexibility to accommodate the variable nature of renewable energy. This could involve investing in demand-side flexibility, power plant flexibility, and energy storage technologies such as pumped-storage hydro and batteries¹. These investments would allow the grid to respond to fluctuations in renewable energy generation and ensure a reliable power supply. - Policymakers could also establish training programs at universities and vocational schools to develop the skilled workforce needed for the renewable energy sector. These programs could focus on the technical skills needed to design, install, and maintain renewable energy systems.
 Enhance energy security and promote cleaner fossil fuel technologies	- Policymakers could focus on increasing domestic production of coal, gas, and petroleum to meet the growing demand and reduce dependence on imports. This could be achieved by providing incentives for exploration and production activities, and by streamlining regulatory processes. - The government could also invest in technologies to make the use of fossil fuels cleaner and more efficient. This could include funding for research into carbon capture and storage technologies, and the development of cleaner burning technologies for coal and gas. - Policymakers could also consider implementing a carbon pricing mechanism to reflect the true cost of fossil fuel use. This would provide an economic incentive for producers and consumers to reduce their carbon emissions.
 Reform the power sector and electricity market	- Policymakers could continue advancing grid modernization efforts by creating an enabling policy environment for reliable and efficient electricity distribution. This could include investments in smart grid technologies and infrastructure upgrades. - The government could also implement market reforms to encourage competition and efficiency in the electricity market. This could include measures to break up state-owned electricity monopolies and encourage private-sector participation. - Policymakers could also promote the use of demand response strategies to manage peak demand. This could include incentives for consumers to reduce their electricity use during peak times.

 Continue promoting energy storage and efficiency	- Policymakers could implement policies to encourage the development and deployment of energy storage technologies. This could include tax incentives for energy storage projects and regulations that require utilities to include energy storage in their resource planning. - Policymakers could also provide incentives for energy efficiency improvements in the industrial sector. This could include tax credits for energy-efficient equipment and funding for energy audits.
 Ensure access and usage of high-quality, modern energy sources	- Policymakers could consider developing state-level plans to ensure universal access to reliable and affordable energy. This plan could outline specific targets for continuous and affordable rural electricity usage and the expansion of clean cooking solutions. - The government could also implement policies to support decentralized renewable energy systems in rural areas. This could include commercial and consumer-facing incentives for solar home systems and mini-grids. - Policymakers could also provide subsidies for energy-efficient appliances for low-income households. These subsidies could help to make these appliances more affordable and reduce energy consumption.
 Promote integrated energy assessments and enhance coordination in energy policymaking	- Policymakers could establishing coordination mechanisms (such as inter-ministerial working committees) responsible for integrated energy assessment and policymaking. Such platforms could coordinate energy policy across different government departments and ensure that energy policy is aligned with national development goals. - The government could also develop integrated resource planning processes that consider all energy resources. This could include a comprehensive assessment of the country's energy resources and an analysis of different energy scenarios.
 Encourage wider adoption of CCUS and create conducive ecosystem for carbon credit trading	- Policymakers could implement a comprehensive policy framework that provides incentives for companies to adopt carbon capture technologies. - Private investment in carbon projects can be attracted by encouraging public-private partnerships. This can be done by providing regulatory support and risk guarantees for private sector investments and new forms of capital such as green bonds. - Policymakers could also implement policies to support negative emission technologies, including funding for research and development in these areas and regulations that encourage the deployment of these technologies. - The government can implement a robust and reliable carbon credit trading market in the country to enable more investment in cleaner technologies.

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